Toward resolvent-based estimation and control of wavepackets in supersonic turbulent jets

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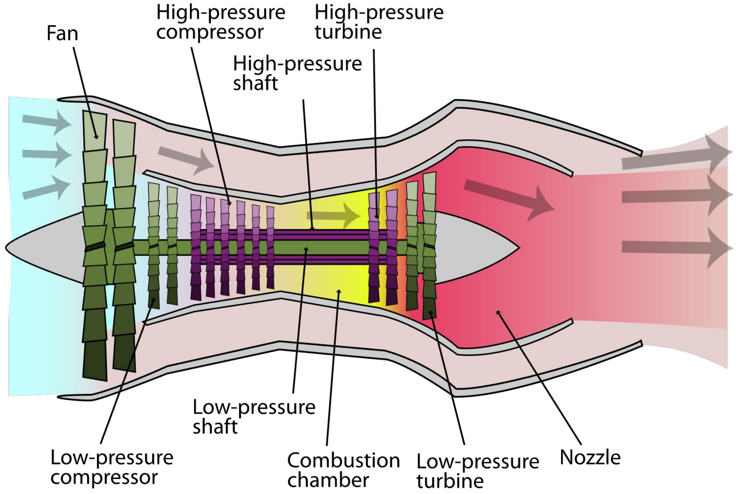
AIAA Scitech Forum 2026

Date: Jan 15, 2026

Motivation: jet noise

- **Commercial applications:**
  - Worldwide dependence on air travel is consistently increasing
  - Strict regulation of commercial aircraft noise enforced by Federal Aviation Administration (FAA)



(Aainsqatsi 2020)

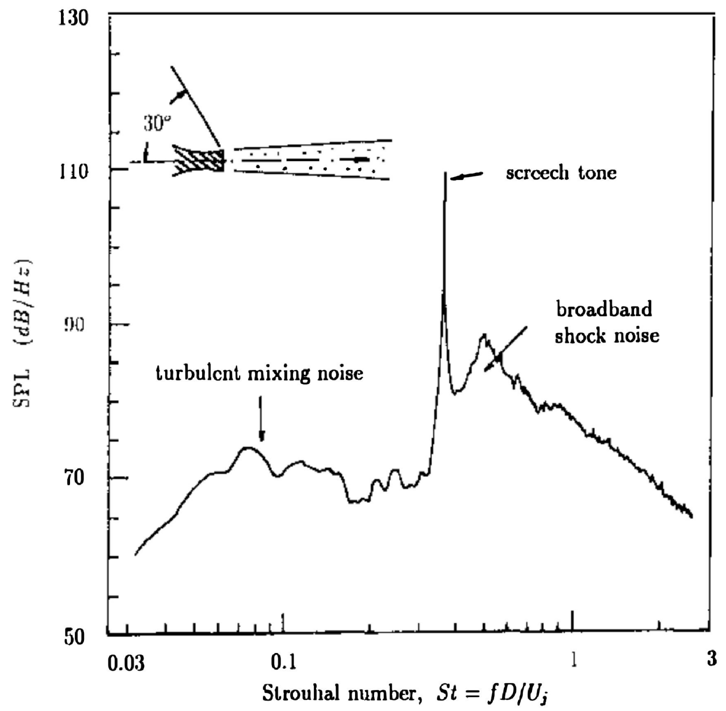
- **Military applications:**
  - Noise levels on Navy aircraft carriers can reach over 150 dB !
  - The U.S. Department of Veteran Affairs spends over $1 billion per year on hearing loss cases (NRAC 2009)

(Aquatical. 2023)

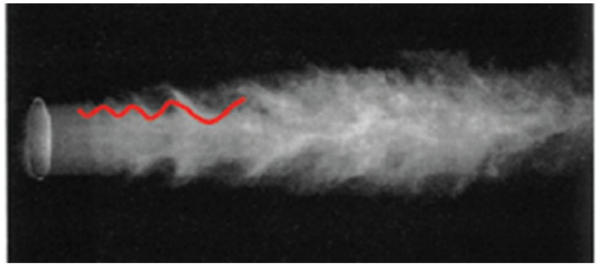
Wavepackets

- **Turbulent mixing noise**
  - High-speed jet exhaust interacts with ambient air, radiates sound wave
  - Dominant source of noise in the downstream direction, where noise levels are highest

- The loudest components of turbulent mixing noise are generated by **wavepackets** (Jordan. 2013)
  - Large-scale organized motions within the jet
  - Space-time coherence
  - Efficient noise sources (Crighton 1990)



(Christopher 1995)



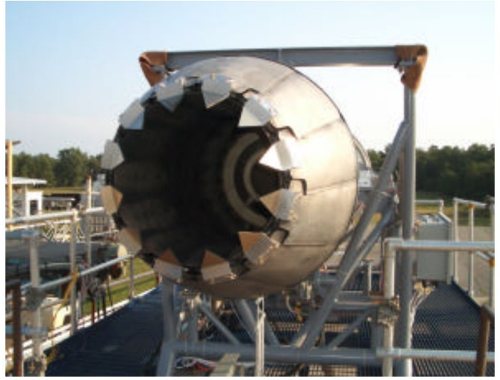
Review: previous efforts on jet noise control

1. Passive control

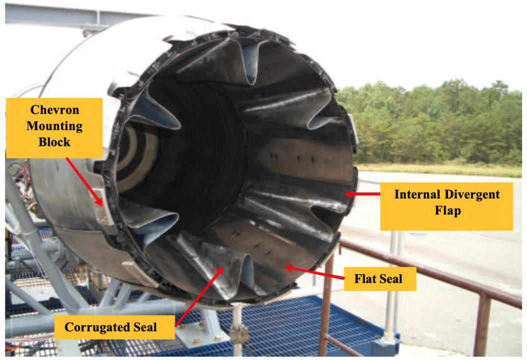
- Chevrons (J. Bridges 2004) and Nozzle inserts (Seiner 2006)

2. Active open loop control

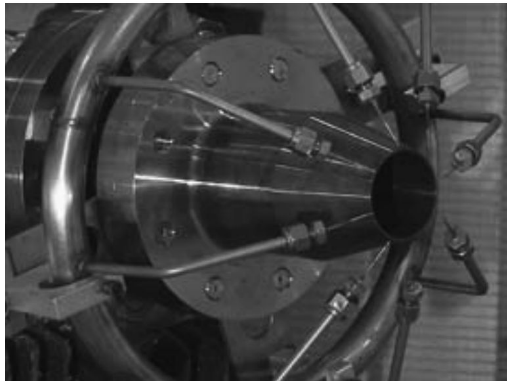
- Steady or periodic micro-jets (Krothapalli 2003,) or plasma actuators (Samimy 2007, 2012) (Sinha 2017)



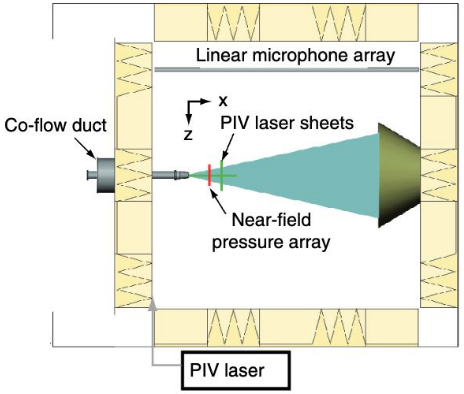
chevrons



inserts



micro-jets



plasma actuators

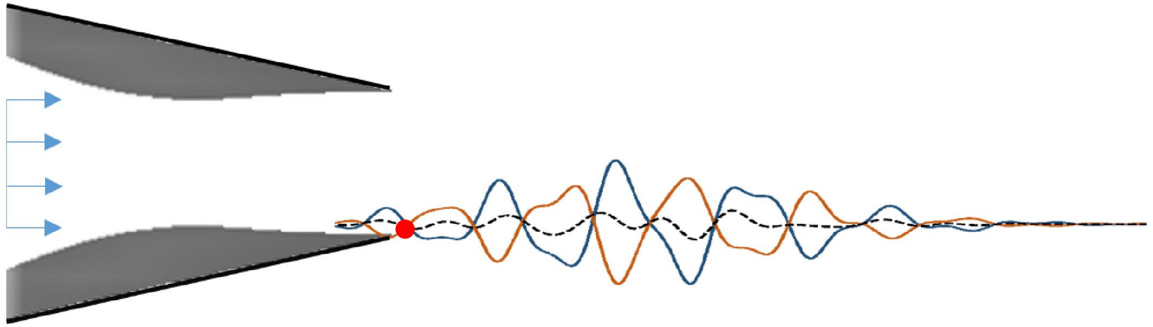
- Both of them achieved modest noise reduction
- Previous approaches reduces noise by indirectly targeting wavepackets
- Therefore, why not directly target noise-producing wavepacktets?

Research Goal & Objective

Objective:

Enable and evaluate a **wavepacket cancellation jet noise mitigation strategy** using a state-of-the-art **estimation and control framework.**

Use near-nozzle actuation to create a opposite wavepacket to linearly cancel the natural wavepacket



System definition

Linear-time invariant system

$$\left.\begin{aligned}\frac{\mathrm{d}\boldsymbol{u}}{\mathrm{d}t}(t) &= \boldsymbol{A}\boldsymbol{u}(t) + \boldsymbol{B}_{\boldsymbol{f}}\boldsymbol{f}(t) + \boldsymbol{B}_{\boldsymbol{a}}\boldsymbol{a}(t),\\ \boldsymbol{y}(t) &= \boldsymbol{C}_{\boldsymbol{y}}\boldsymbol{u}(t) + \boldsymbol{n}(t),\\ \boldsymbol{z}(t) &= \boldsymbol{C}_{\boldsymbol{z}}\boldsymbol{u}(t),\end{aligned}\right\}$$

Forcing-driven system

$$\left.\begin{aligned}\frac{\mathrm{d}\boldsymbol{u}_1}{\mathrm{d}t}(t) &= \boldsymbol{A}\boldsymbol{u}_1(t) + \boldsymbol{B}_{\boldsymbol{f}}\boldsymbol{f}(t)\\ \boldsymbol{y}_1(t) &= \boldsymbol{C}_{\boldsymbol{y}}\boldsymbol{u}_1(t) + \boldsymbol{n}(t)\\ \boldsymbol{z}_1(t) &= \boldsymbol{C}_{\boldsymbol{z}}\boldsymbol{u}_1(t)\end{aligned}\right\}$$

Actuation-driven system

$$\left.\begin{aligned}\frac{\mathrm{d}\boldsymbol{u}_2}{\mathrm{d}t}(t) &= \boldsymbol{A}\boldsymbol{u}_2(t) + \boldsymbol{B}_a\boldsymbol{a}(t),\\ \boldsymbol{y}_2(t) &= \boldsymbol{C}_y\boldsymbol{u}_2(t)\\ z_2(t) &= \boldsymbol{C}_z\boldsymbol{u}_2(t)\end{aligned}\right\}$$

Resolvent-based estimation:

Forcing-driven system

$$\left.\begin{aligned} \frac{\mathrm{d}\boldsymbol{u}_1}{\mathrm{d}t}(t) &= \boldsymbol{A}\boldsymbol{u}_1(t) + \boldsymbol{B}_{\boldsymbol{f}}\boldsymbol{f}(t) \\ \boldsymbol{y}_1(t) &= \boldsymbol{C}_{\boldsymbol{y}}\boldsymbol{u}_1(t) + \boldsymbol{n}(t) \\ \boldsymbol{z}_1(t) &= \boldsymbol{C}_{\boldsymbol{z}}\boldsymbol{u}_1(t) \end{aligned}\right\}$$

Estimator

$$\tilde{\boldsymbol{z}}(t) = \int_{-\infty}^{\infty} \boldsymbol{T}_z(\tau)\boldsymbol{y}(t-\tau)\mathrm{d}\tau \quad \overset{\text{FT}}{\Longleftrightarrow} \quad \hat{\tilde{z}}(\omega) = \hat{\boldsymbol{T}}_z\hat{\boldsymbol{y}}(\omega)$$

Error

$$e(t) = z(t) - \tilde{z}(t)$$

Cost function

$$J = \int_{-\infty}^{\infty} \left\langle \boldsymbol{e}^\dagger(t)\boldsymbol{e}(t)\right\rangle \mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} \left\langle \hat{\boldsymbol{e}}^\dagger(\omega)\hat{\boldsymbol{e}}(\omega)\right\rangle \mathrm{d}\omega$$

Minimize

Optimal estimator:

$$\hat{\boldsymbol{T}}_z\boxed{\hat{\boldsymbol{G}}_{\boldsymbol{I}}} = \boxed{\hat{\boldsymbol{G}}_{\boldsymbol{r}}}$$

$$\hat{\boldsymbol{G}}_I = \boxed{\boldsymbol{R}_{\boldsymbol{y}f}}\left\langle \hat{\boldsymbol{f}}\hat{\boldsymbol{f}}^\dagger\right\rangle \boldsymbol{R}_{yf}^\dagger + \left\langle \hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^\dagger\right\rangle$$

$$\hat{\boldsymbol{G}}_r = \boxed{\boldsymbol{R}_{zf}}\left\langle \hat{\boldsymbol{f}}\hat{\boldsymbol{f}}^\dagger\right\rangle \boldsymbol{R}_{yf}^\dagger$$

$$\boldsymbol{R}_{yf}(\omega) = \boldsymbol{C}_y\boxed{\boldsymbol{R}}\boldsymbol{B}_f$$

$$\boldsymbol{R}_{zf}(\omega) = \boldsymbol{C}_z\boxed{\boldsymbol{R}}\boldsymbol{B}_f$$

$$\boldsymbol{R} = (-\mathrm{i}\omega\boldsymbol{I} - \boldsymbol{A})^{-1}$$

Data-driven estimation:

Optimal estimator:

$$\hat{\boldsymbol{T}}_z \boxed{\hat{\boldsymbol{G}}_{\boldsymbol{I}}} = \boxed{\hat{\boldsymbol{G}}_{\boldsymbol{r}}}$$

$$\begin{bmatrix} \boldsymbol{S}_{y_1,y_1} & \boldsymbol{S}_{y_1,z_1} \\ \boldsymbol{S}_{z_1,y_1} & \boldsymbol{S}_{z_1,z_1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}_{yf} \\ \boldsymbol{R}_{zf} \end{bmatrix} \hat{\boldsymbol{F}} \begin{bmatrix} \boldsymbol{R}^{\dagger}_{\boldsymbol{y}f} & \boldsymbol{R}^{\dagger}_{zf} \end{bmatrix} + \begin{bmatrix} \hat{\boldsymbol{N}} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \boxed{\hat{\boldsymbol{G}}_{\boldsymbol{I}}} & \boxed{\hat{\boldsymbol{G}}_{\boldsymbol{r}}} \\ \hat{\boldsymbol{G}}^{\dagger}_{\boldsymbol{r}} & \boldsymbol{R}_{zf} \hat{\boldsymbol{F}} \boldsymbol{R}^{\dagger}_{zf} \end{bmatrix}$$

(Martini et al. 2022)

Resolvent-based Control:

Actuation-driven system

$$\left.\begin{aligned}\frac{\mathrm{d}\boldsymbol{u}_2}{\mathrm{d}t}(t) &= \boldsymbol{A}\boldsymbol{u}_2(t) + \boldsymbol{B}_a\boldsymbol{a}(t),\\ \boldsymbol{y}_2(t) &= \boldsymbol{C}_y\boldsymbol{u}_2(t)\\ z_2(t) &= \boldsymbol{C}_z\boldsymbol{u}_2(t)\end{aligned}\right\}$$

$$\hat{\boldsymbol{y}}_2(\omega) = \boldsymbol{R}_{ya}(\omega)\hat{\boldsymbol{a}}(\omega), \text{ with } \boldsymbol{R}_{ya}(\omega) = \boldsymbol{C}_y\boldsymbol{R}\boldsymbol{B}_a$$

$$\hat{z}_2(\omega) = \boldsymbol{R}_{za}(\omega)\hat{\boldsymbol{a}}(\omega), \text{ with } \boldsymbol{R}_{za}(\omega) = \boldsymbol{C}_z\boldsymbol{R}\boldsymbol{B}_a$$

Controller

$$\boldsymbol{a}(t) = \int_{-\infty}^{\infty} \boldsymbol{\Gamma}(\tau)\boldsymbol{y}_1(t-\tau)\mathrm{d}\tau, \quad \hat{\boldsymbol{a}} = \hat{\boldsymbol{\Gamma}}\hat{\boldsymbol{y}}_1$$

Cost function

$$J = \int_{-\infty}^{\infty} \langle z^\dagger(t)z(t) + \boldsymbol{a}^\dagger(t)\boldsymbol{P}\boldsymbol{a}(t)\rangle\,\mathrm{d}t = \int_{-\infty}^{\infty} \left\langle \hat{z}^\dagger\hat{\boldsymbol{z}} + \hat{\boldsymbol{a}}^\dagger\boldsymbol{P}\hat{\boldsymbol{a}}\right\rangle \mathrm{d}\omega$$

Minimize

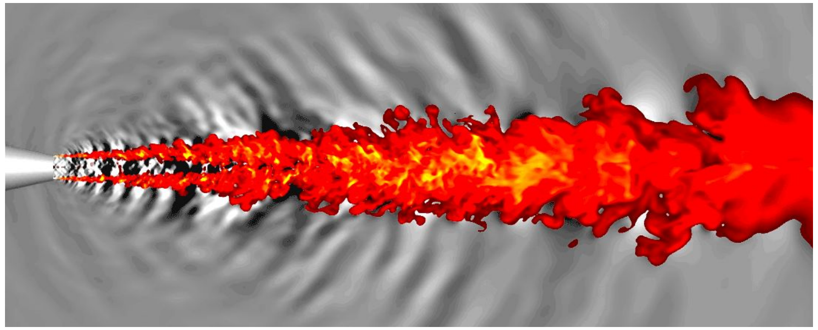
Optimal Controller

$$\hat{\boldsymbol{H}}_l\hat{\boldsymbol{\Gamma}}\hat{\boldsymbol{G}}_I = \hat{\boldsymbol{H}}_r\hat{\boldsymbol{G}}_r$$

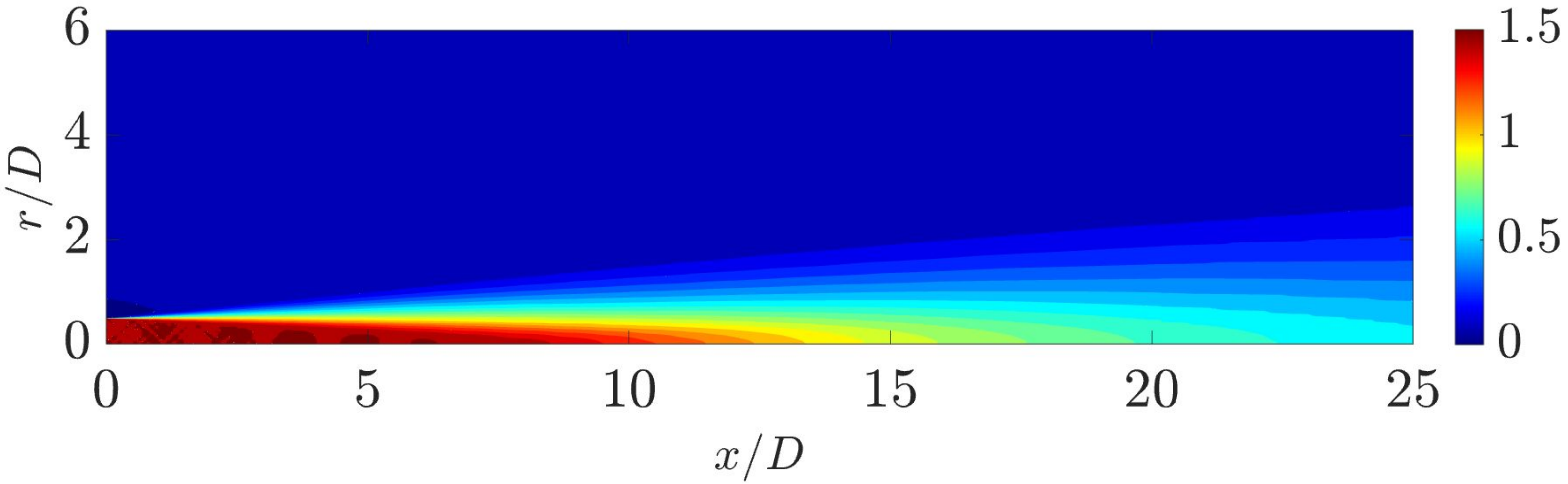
$$\hat{\boldsymbol{H}}_l(\omega) = \boldsymbol{R}_{za}^\dagger(\omega)\boldsymbol{R}_{za}(\omega) + \boldsymbol{P}, \quad \hat{\boldsymbol{H}}_r(\omega) = -\boldsymbol{R}_{za}^\dagger(\omega)$$

Large Eddy Simulation:

- To perform data-driven estimation and mean flow extraction
- Simulations are performed on DoD Onyx and Carpenter platform.

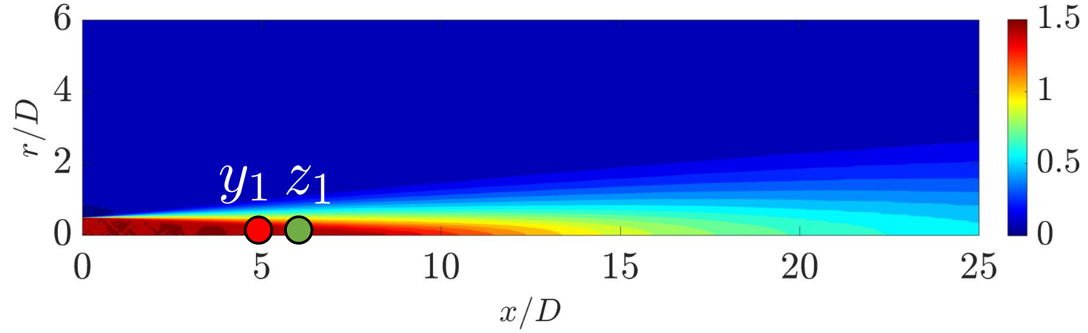


Isothermal, ideally expanded jet

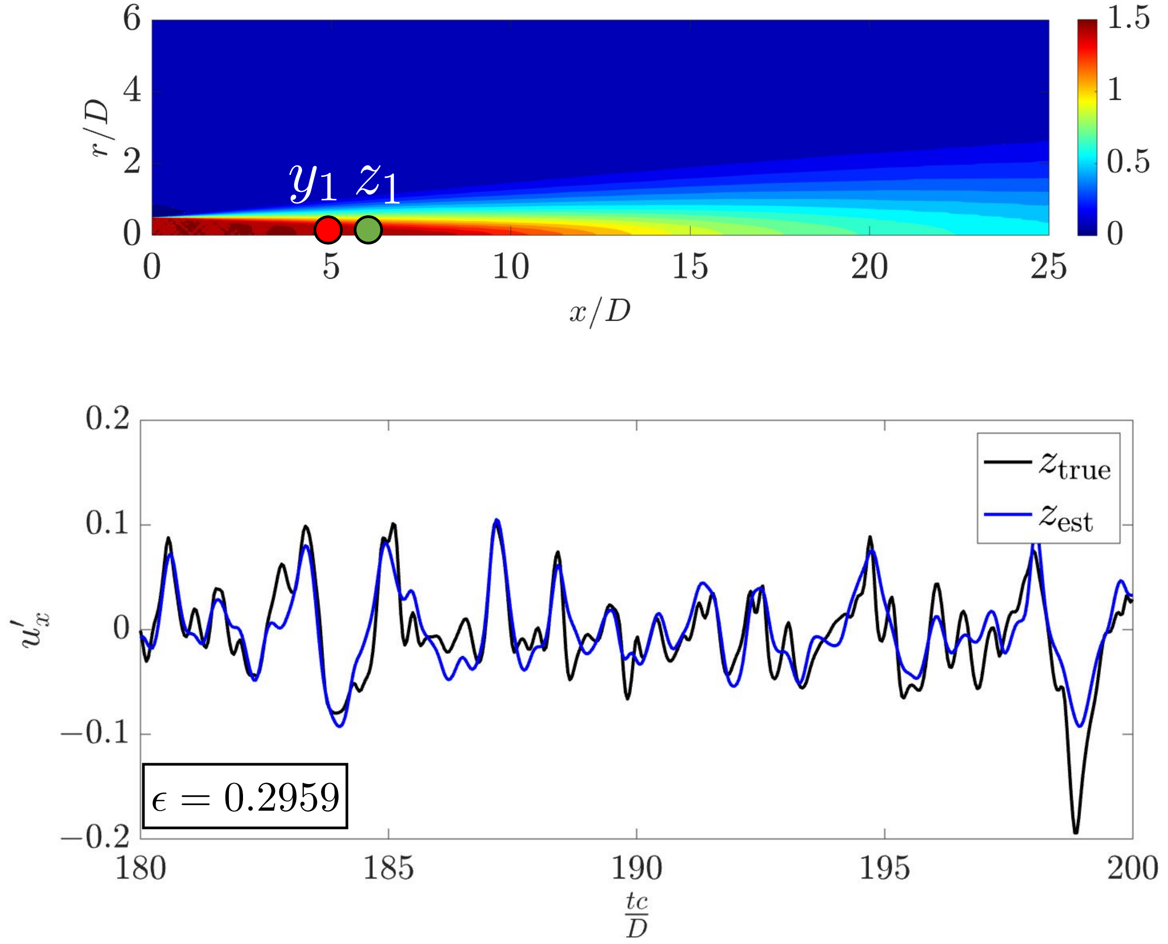


Streamwise mean velocity profile

Date-driven estimation: single-sensor single-target



Date-driven estimation: single-sensor single-target



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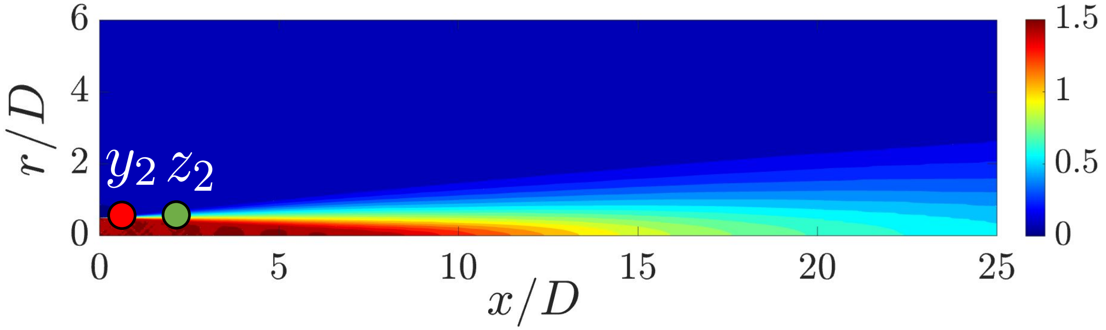

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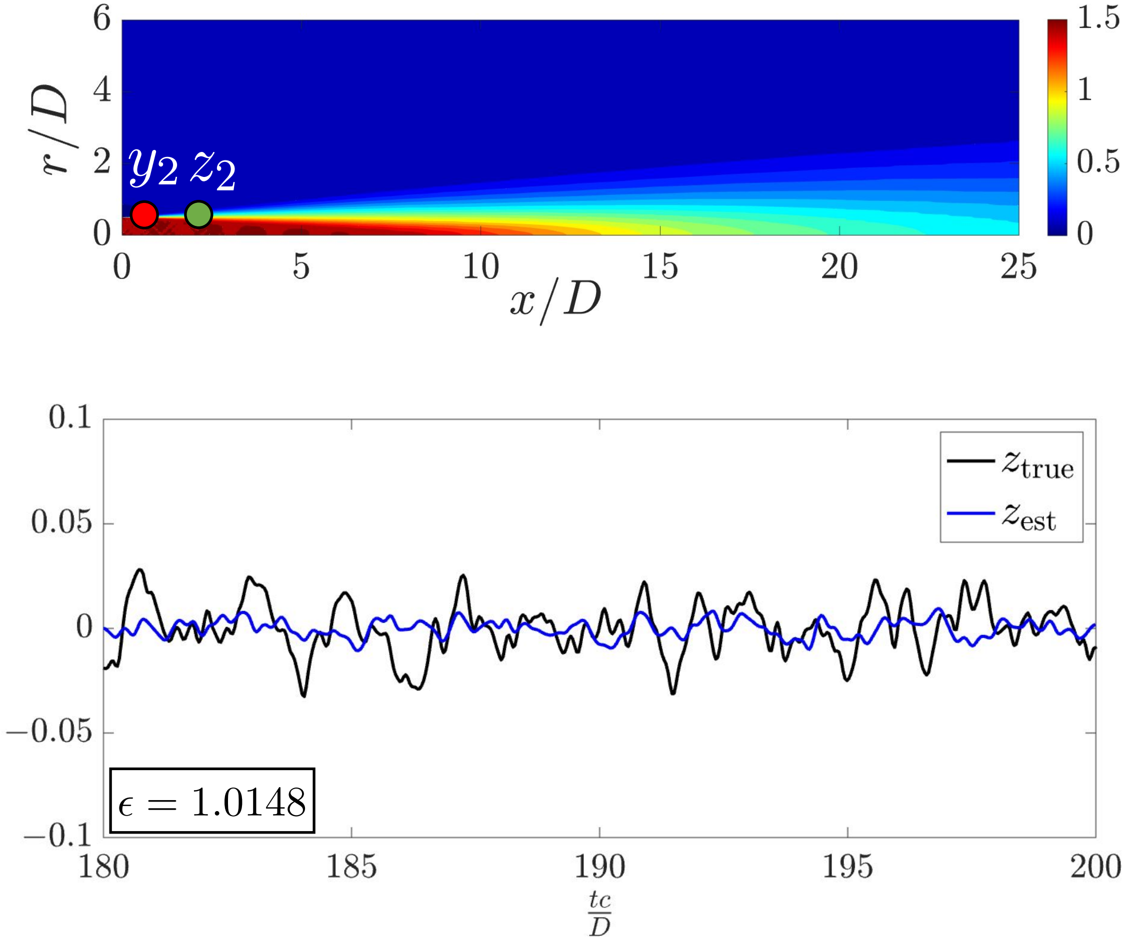
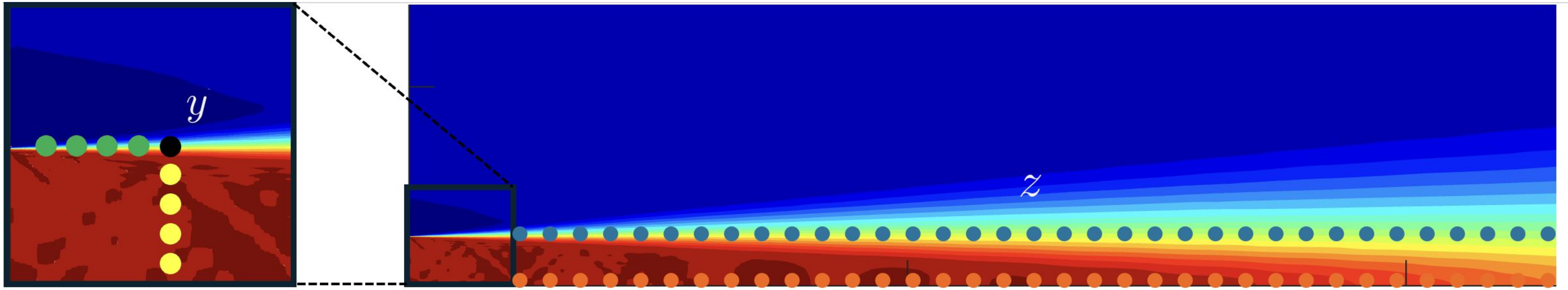


Data-driven estimation: multi-sensor multi-target



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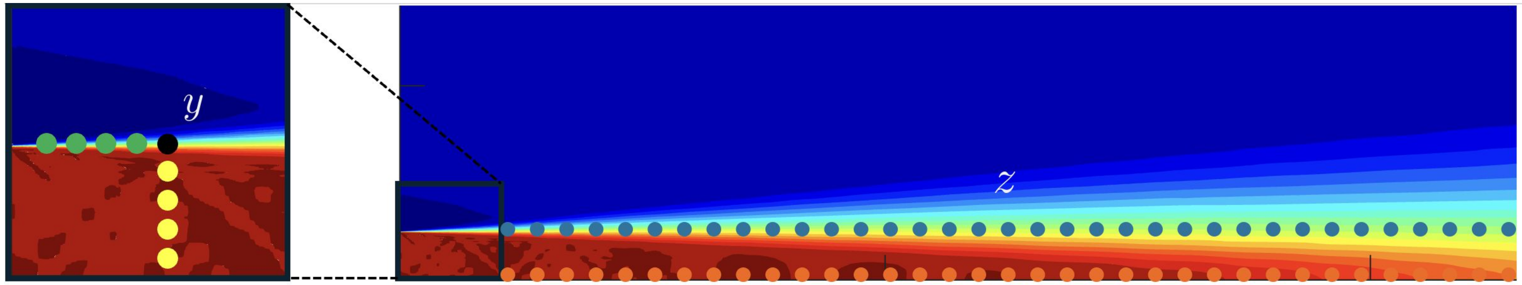

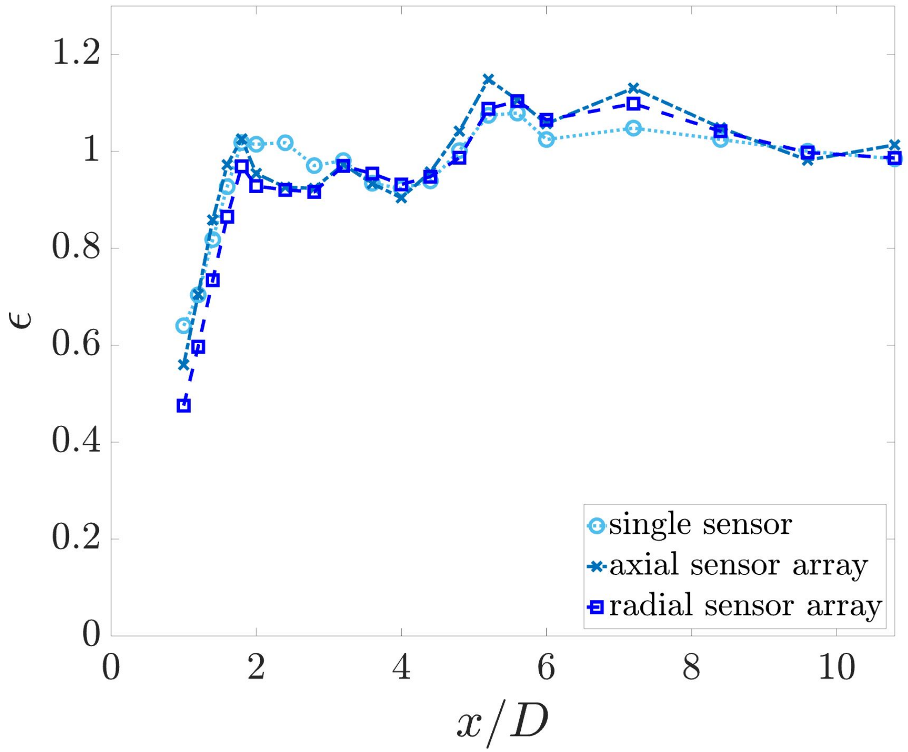

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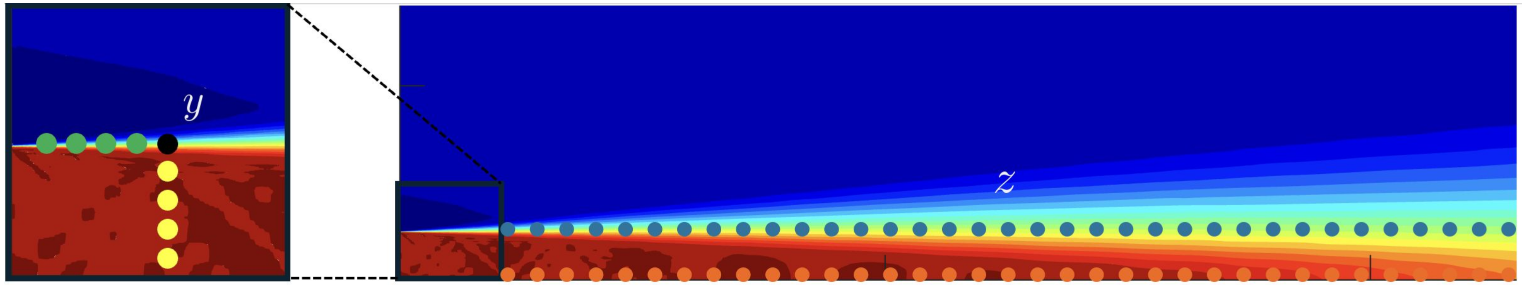
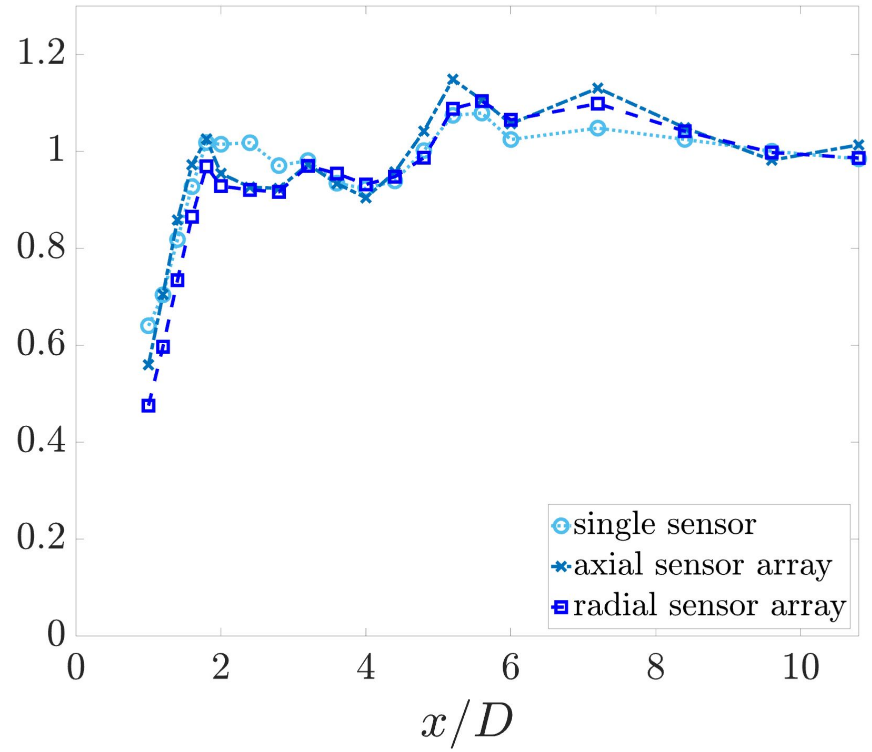
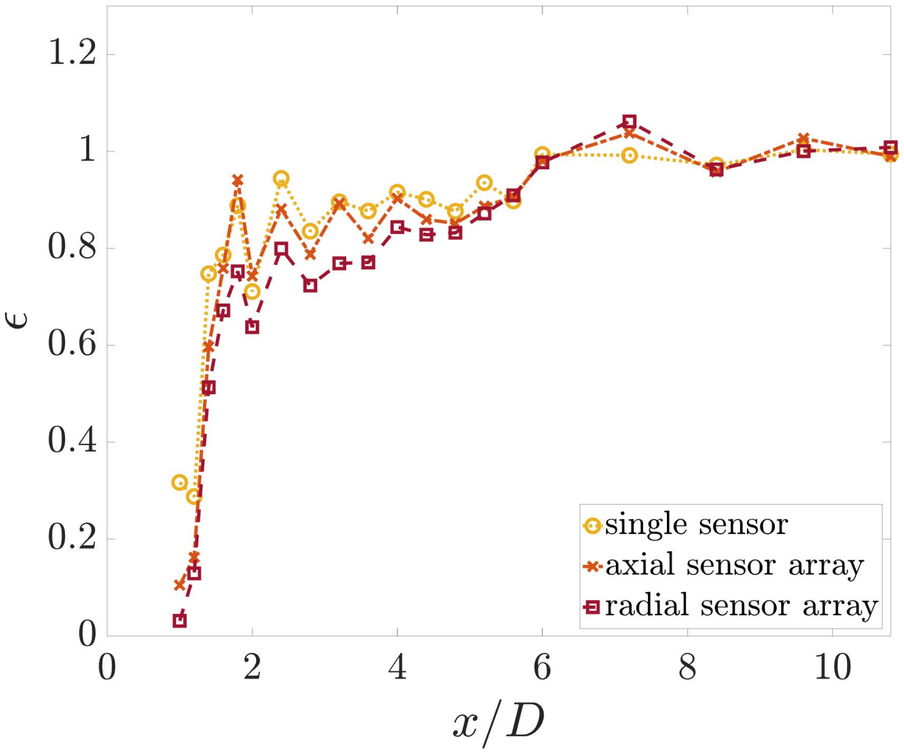


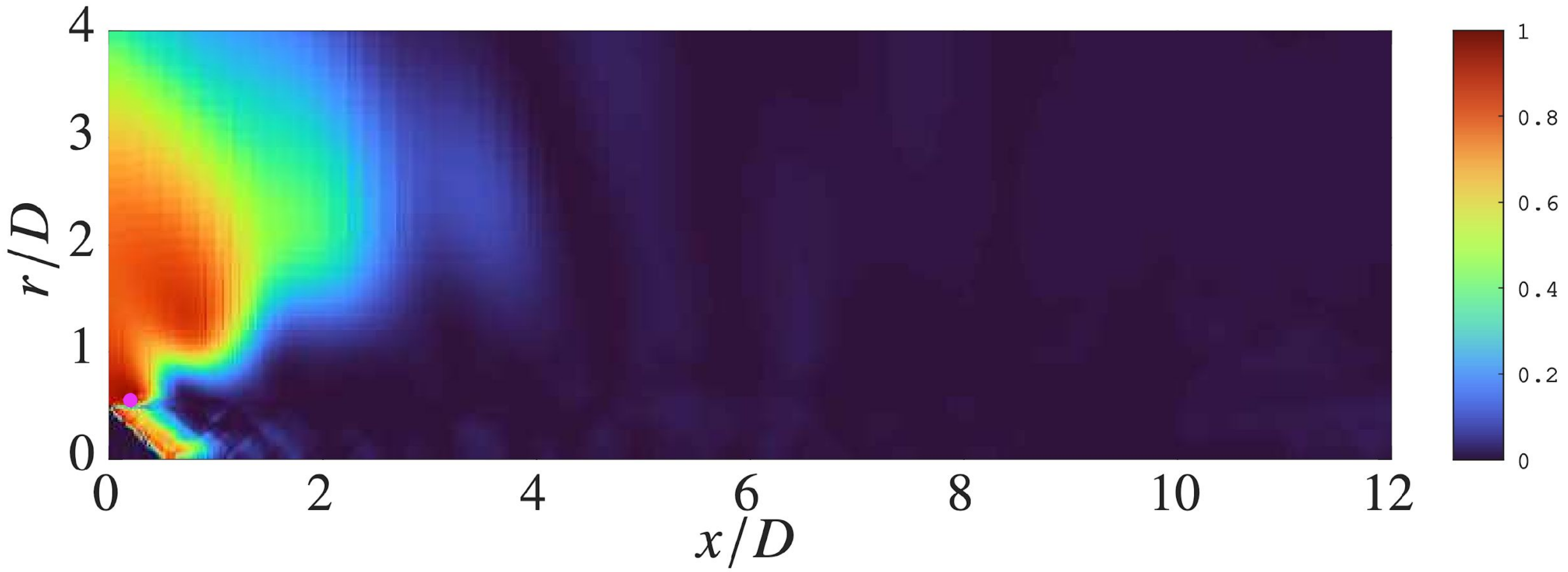




Data-driven estimation: coherence

$$\gamma_{yz} = \frac{|S_{yz}|}{\sqrt{S_{yy}}\sqrt{S_{zz}}}$$

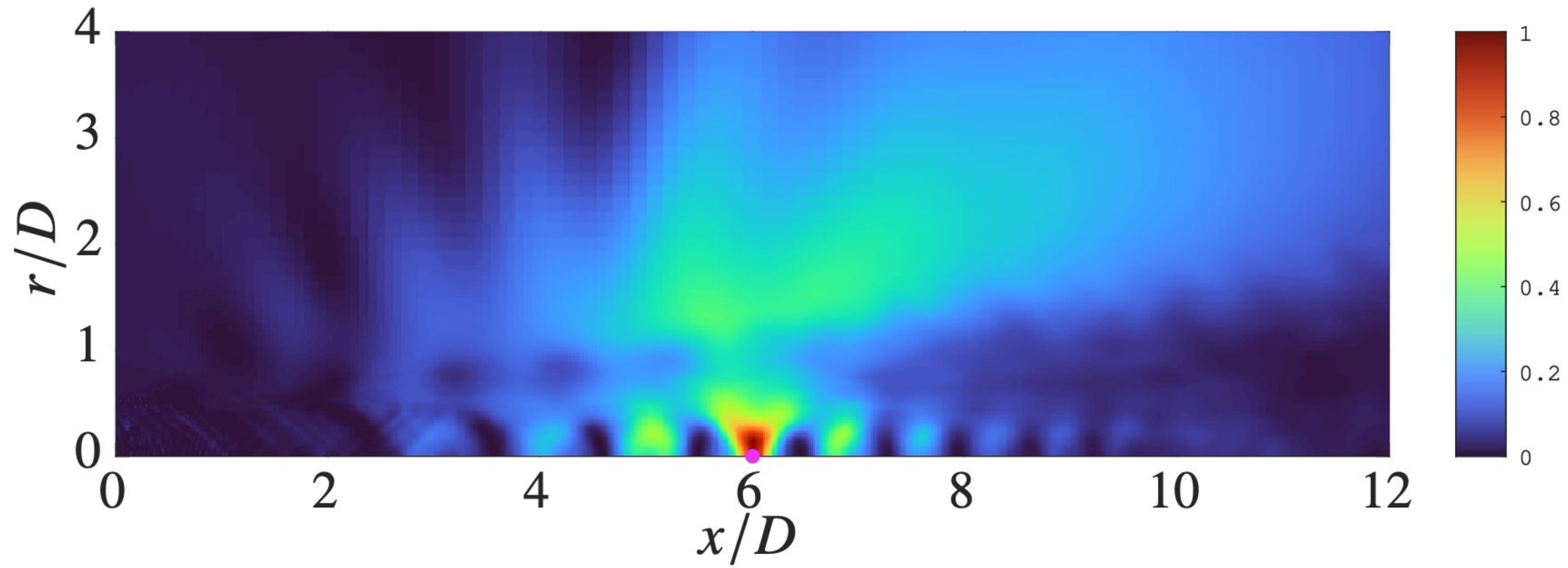
$y : (x/D, r/D) = (0.2, 0.5)$



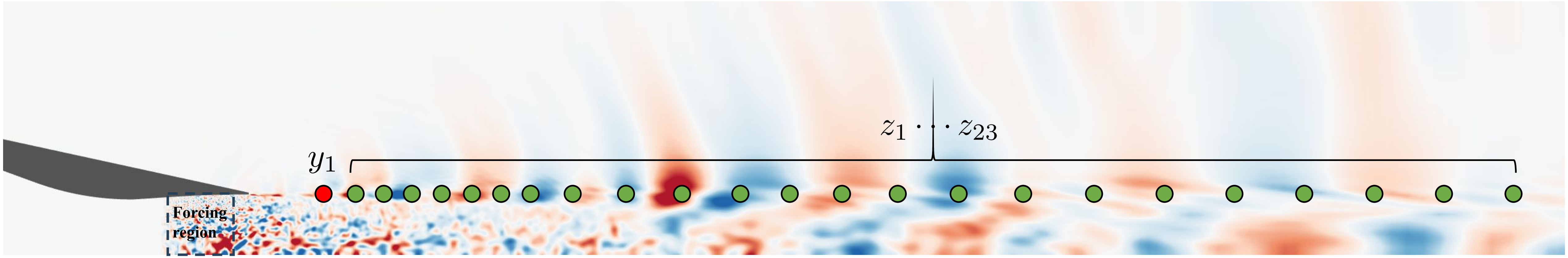
Data-driven estimation: coherence

$$\gamma_{yz} = \frac{|S_{yz}|}{\sqrt{S_{yy}}\sqrt{S_{zz}}}$$

$z : (x/D, r/D) = (6.0, 0)$



Resolvent-based estimation of the linearized jet



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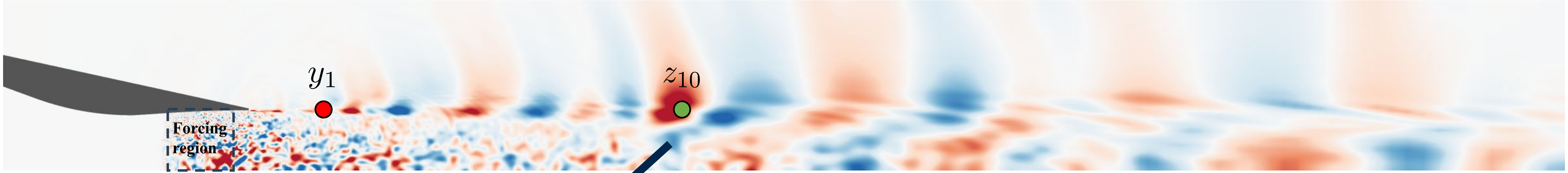

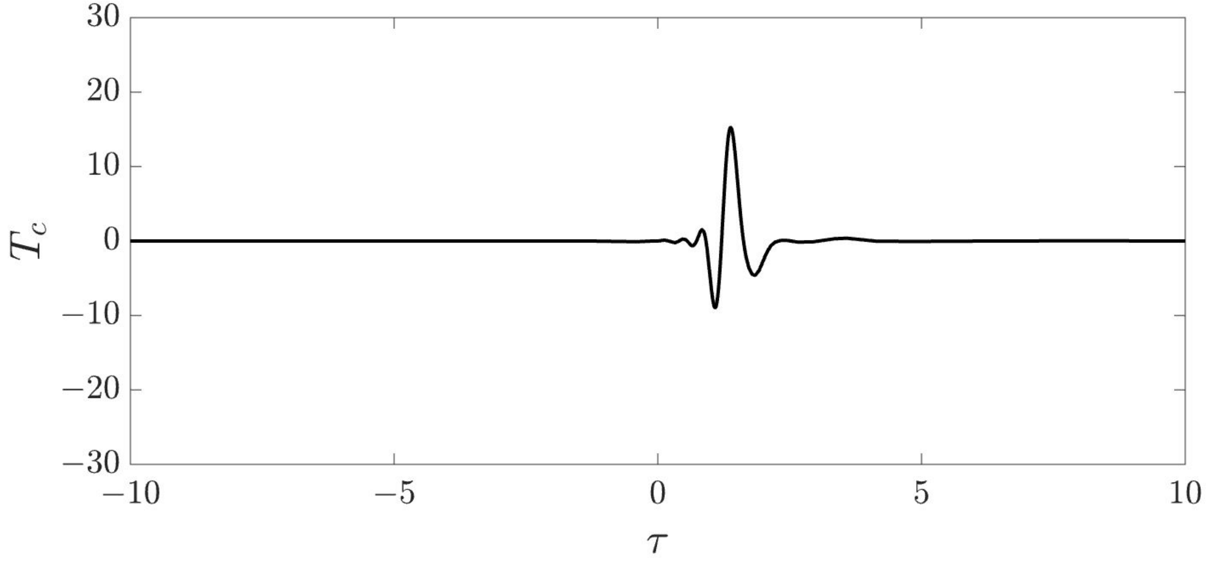

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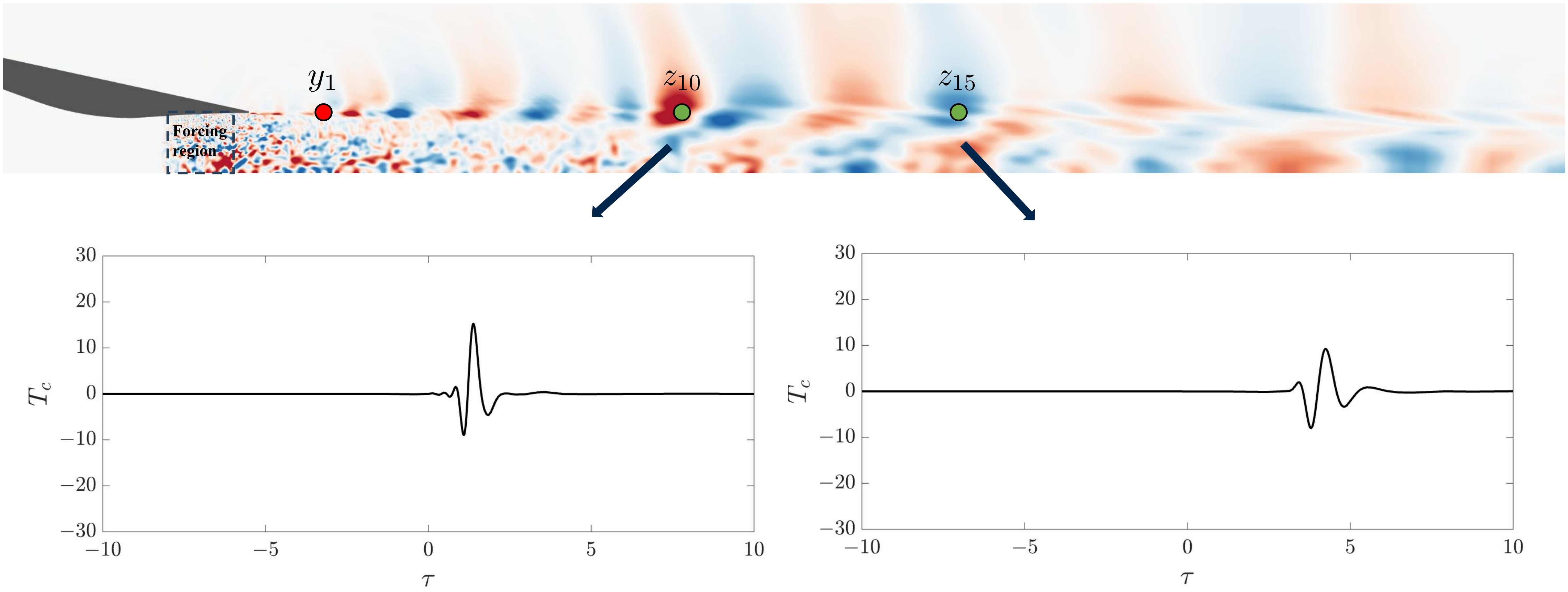

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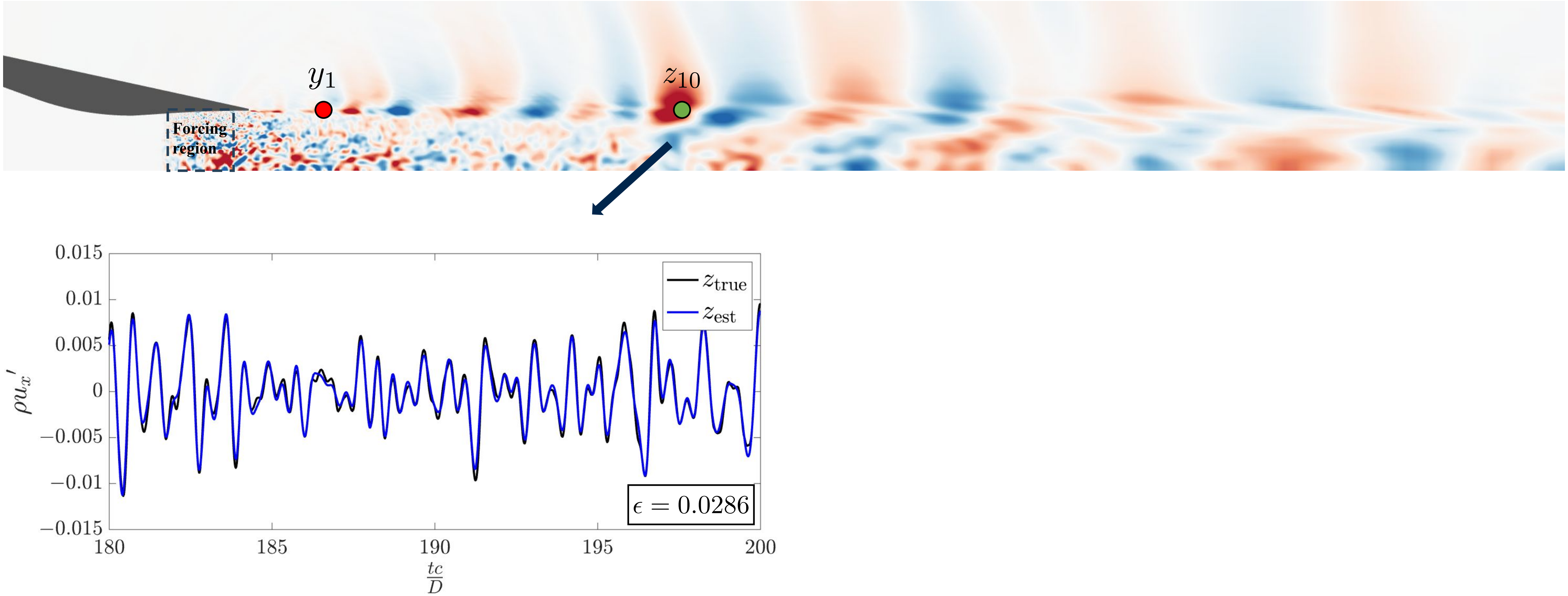

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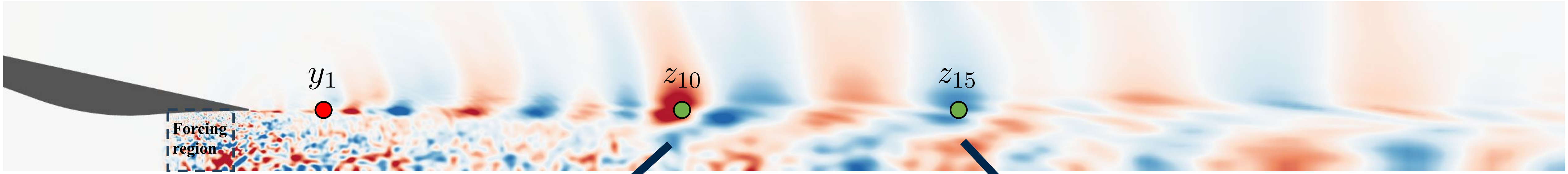

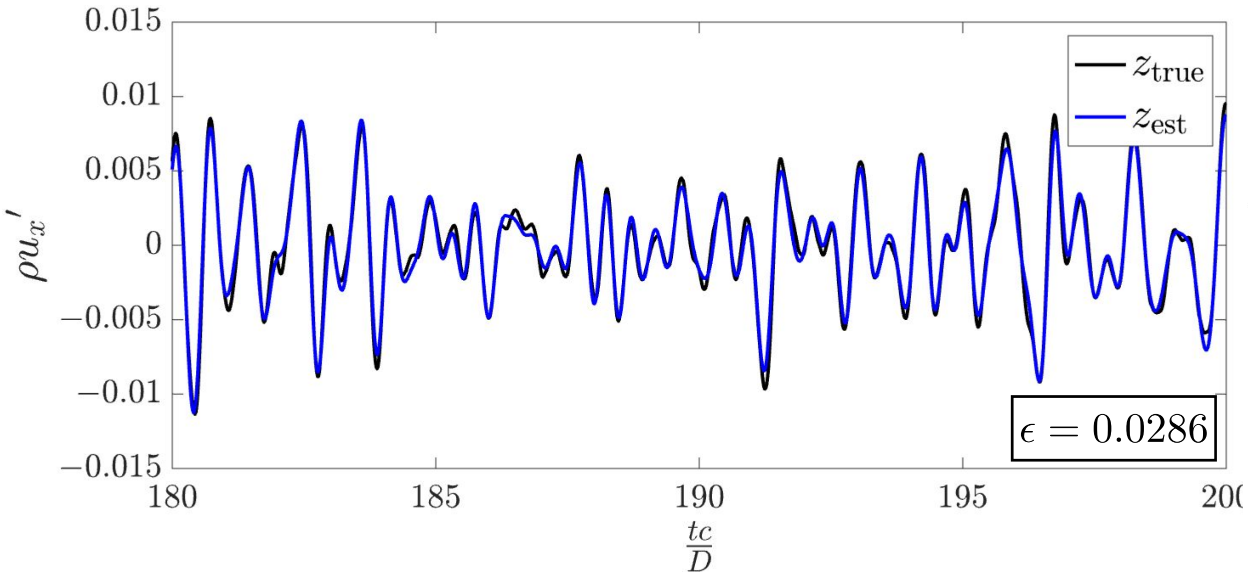

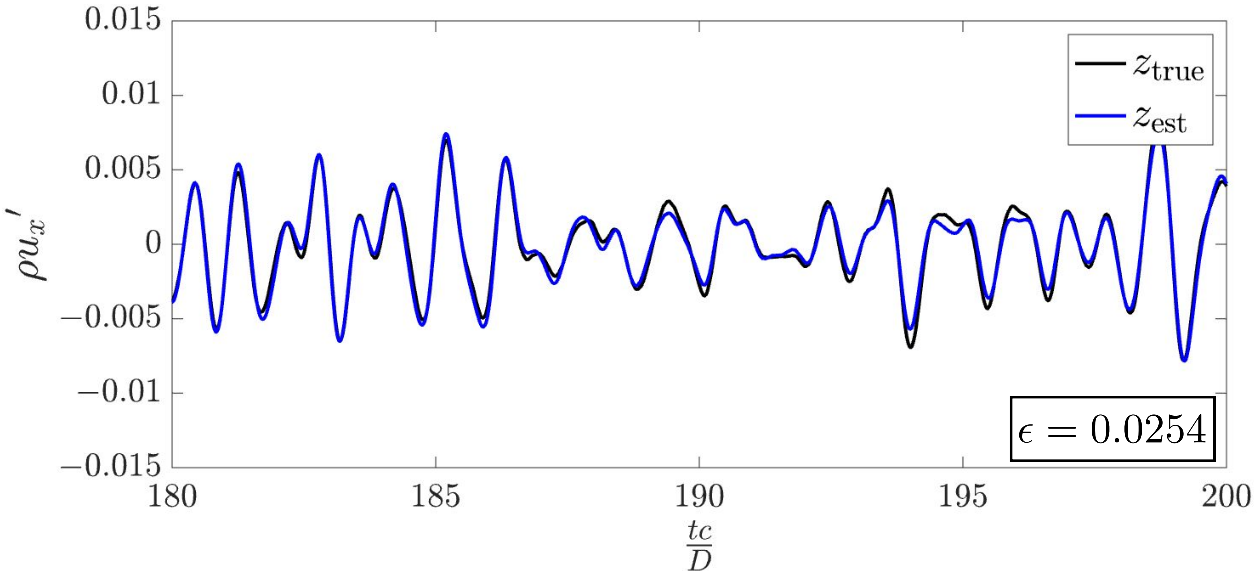

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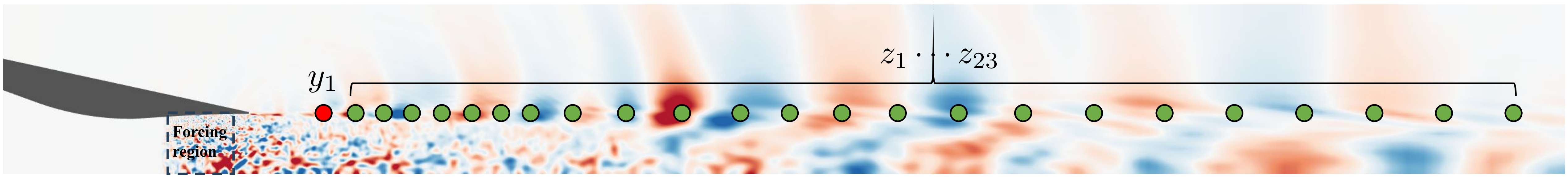
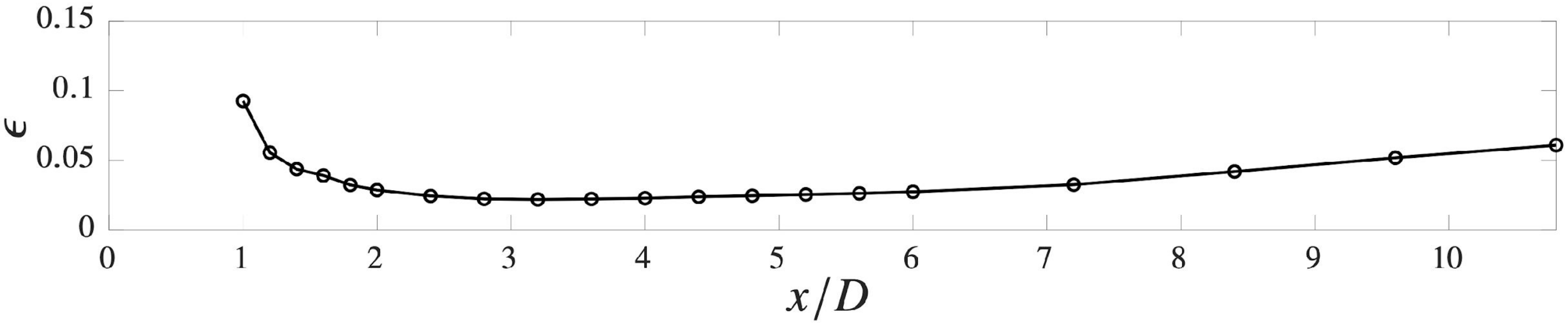
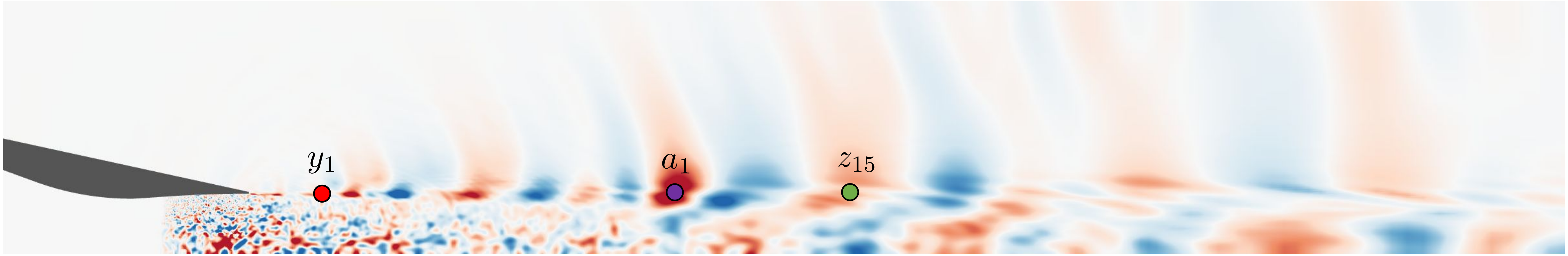


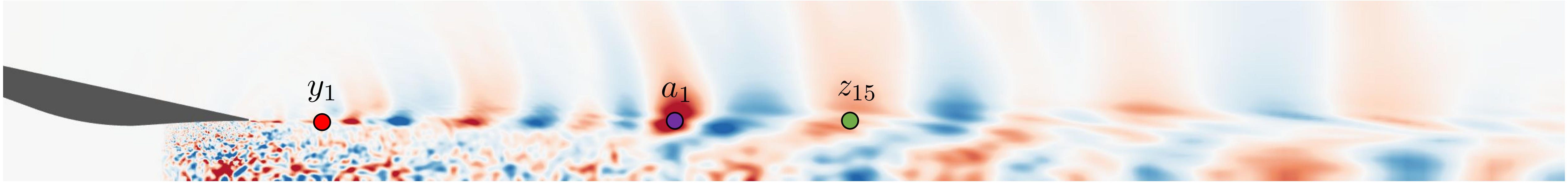


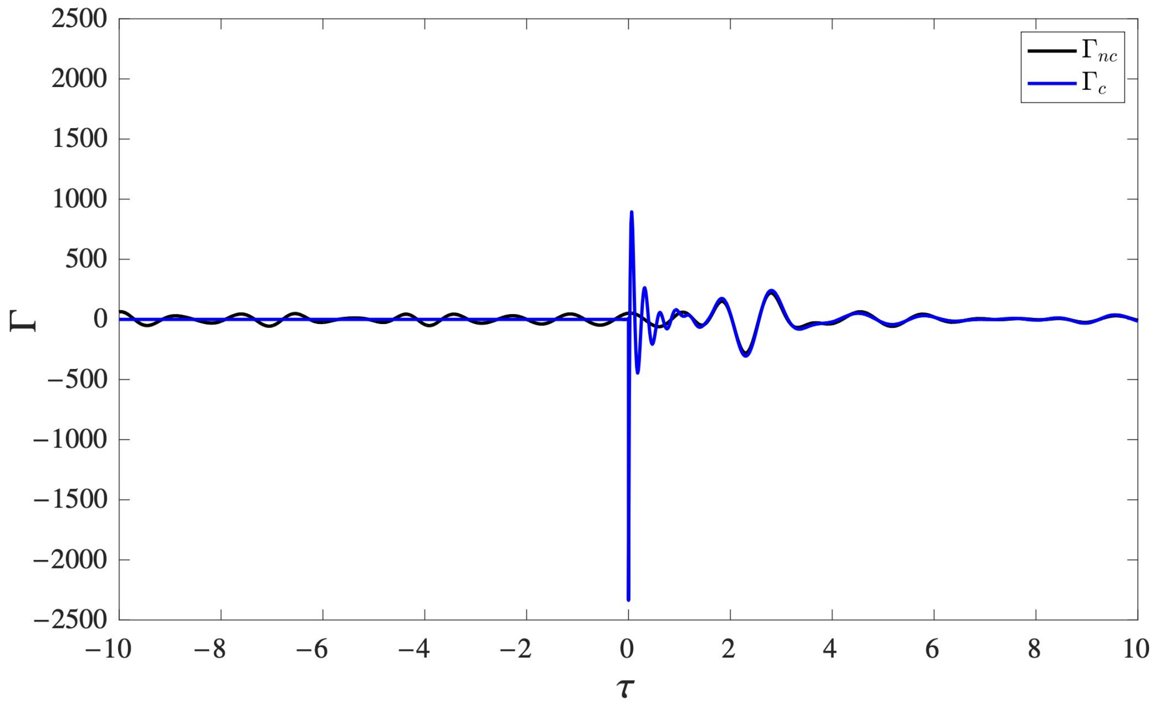
Resolvent-based control of the linearized jet



$$y_1 : (x/D, r/D) = (0.6, 0.5)$$
$$a_1 : (x/D, r/D) = (3.0, 0.5)$$
$$z_{15} : (x/D, r/D) = (5.2, 0.5)$$

Resolvent-based control of the linearized jet





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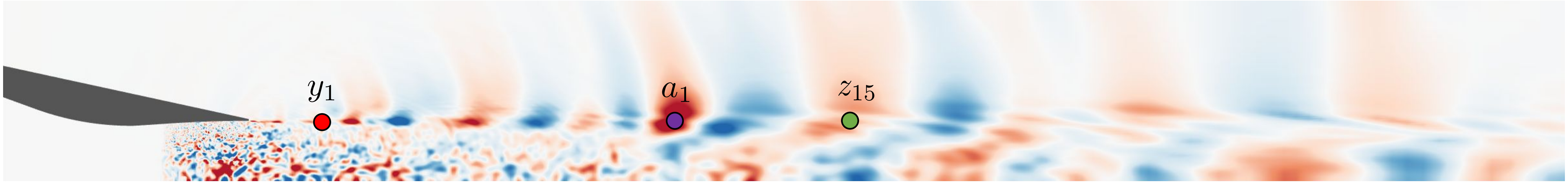

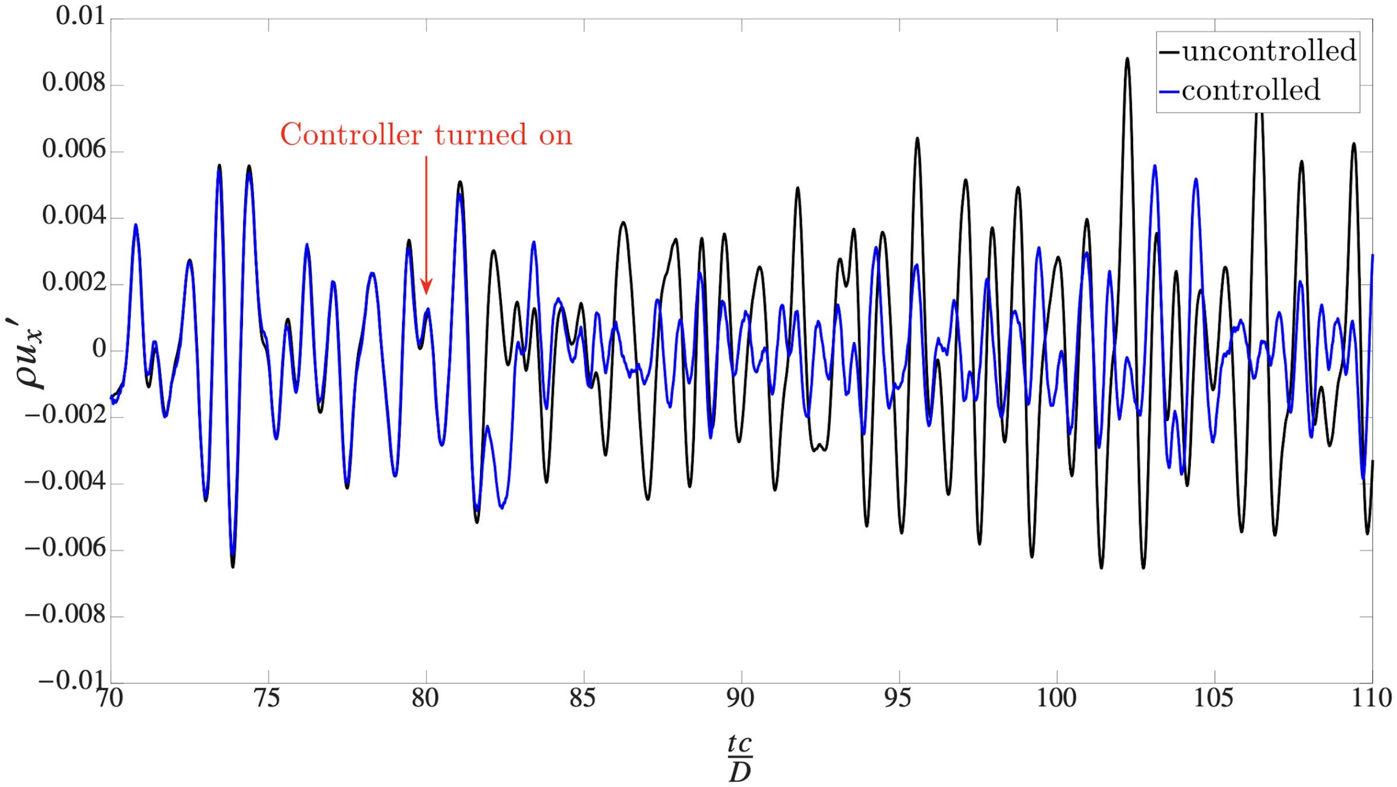

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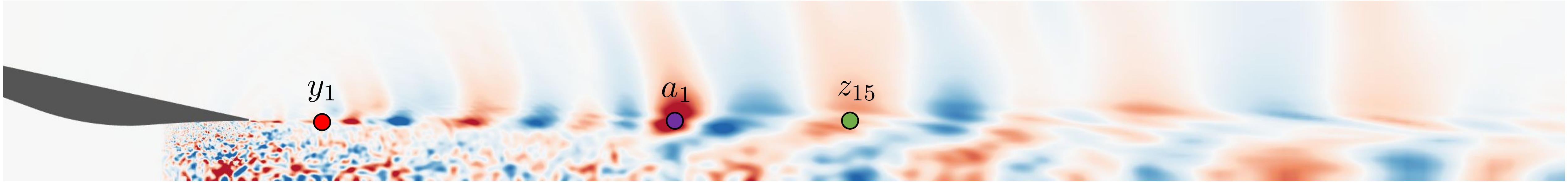

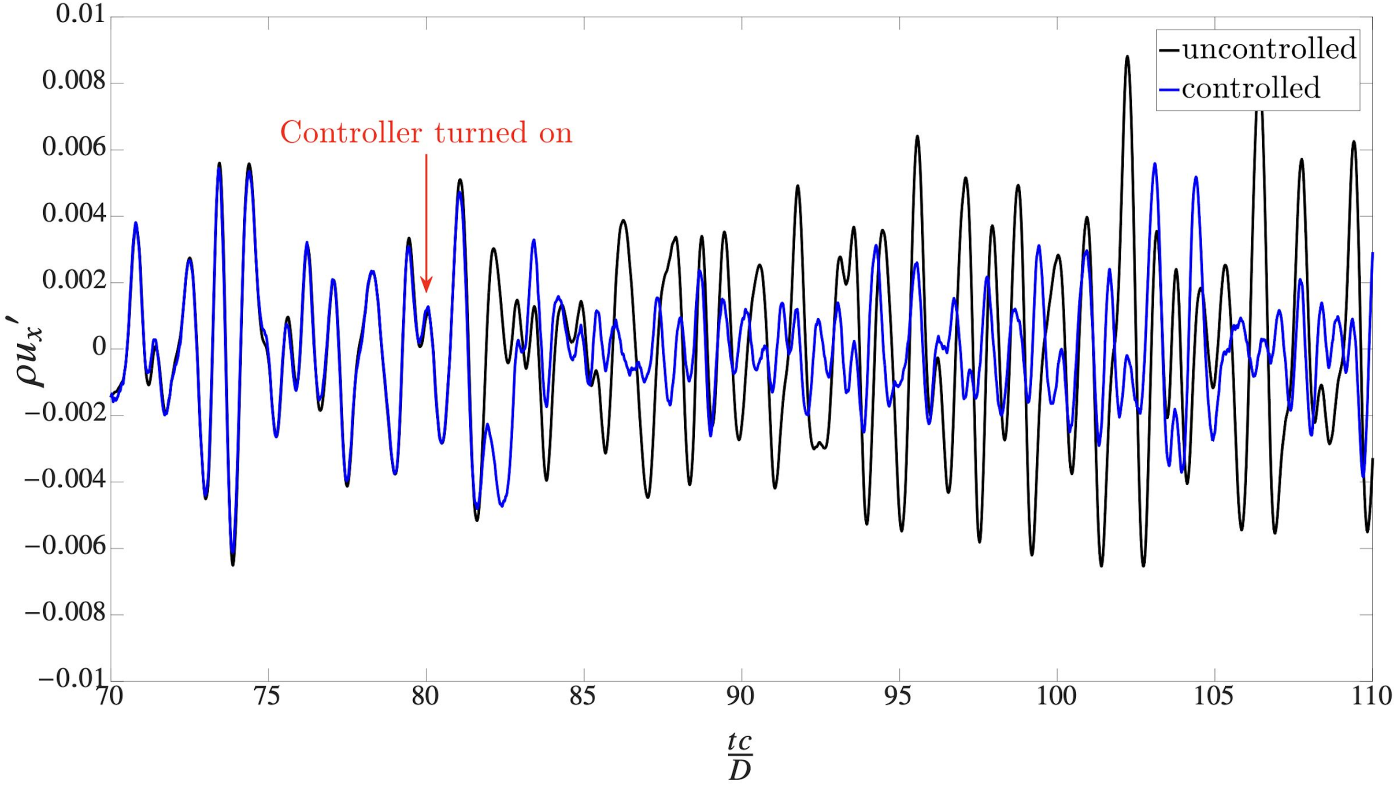

$$(\rho u'_{\mathrm{unctrl}})_{\mathrm{RMS}} = 0.37$$

$$(\rho u'_{\mathrm{ctrl}})_{\mathrm{RMS}} = 0.24$$

34% decrease (averaged over 270 acoustics time units)

Conclusion and future work

- Conclusion
  - Data-driven estimation using LES shows better accuracy for downstream sensors and targets, while accuracy is reduced near the nozzle due to strong nonlinearity and shock–turbulence interactions.
  - Deployed a resolvent-based approach for estimating and controlling downstream wavepacket structures and validated it using a linearized supersonic jet; a 34% reduction in RMS streamwise momentum perturbation is achieved.
  - Results indicate that this method is capable of predicting and mitigating downstream wavepackets, showing promise for reducing turbulent mixing noise in supersonic jets.
- Future work
  - Develop/apply improved models or methods to address near-nozzle nonlinearity.
  - Integrate the control framework into LES jet simulations.

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  - Develop/apply improved models or methods to address near-nozzle nonlinearity.
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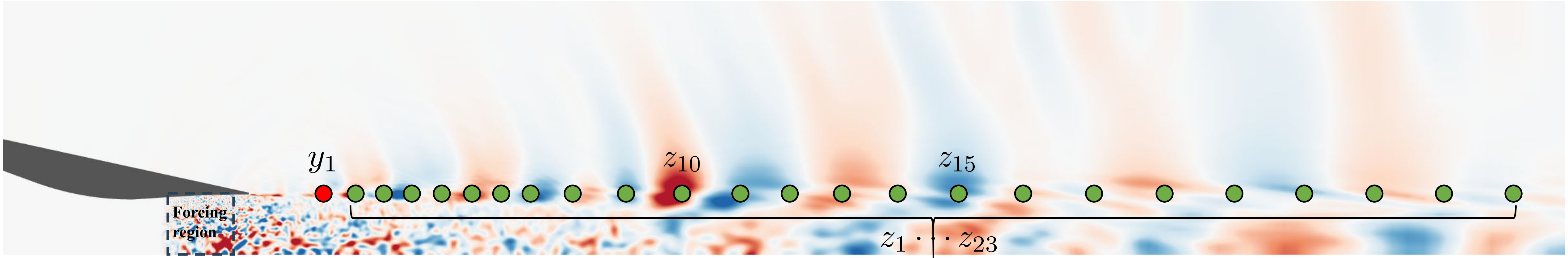
Thank you!

Funded by ONR: N00014-22-2561

Resolvent-based estimation of the linearized jet



$y_1 : (x/D, r/D) = (0.6, 0.5)$

$z_{10} : (x/D, r/D) = (3, 0.5)$

$z_{15} : (x/D, r/D) = (5.2, 0.5)$

$z_1 \sim z_{23} : (x/D, r/D) = (1.0 \ \sim 10.0, 0.5)$